Real Time System

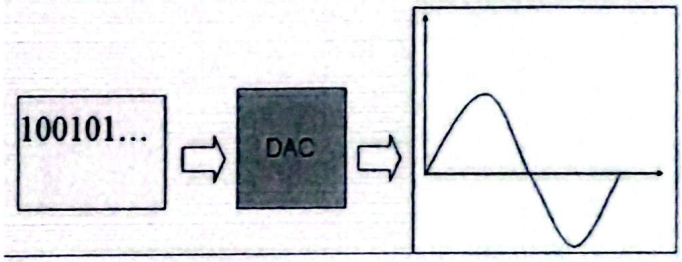
Third Level

Lecture Three

Digital to Analog Converters (DAC)

Digital to Analog Converters

A digital to analog converter (DAC) converts a digital signal or values to an analog voltage or current output.



Types of DACs

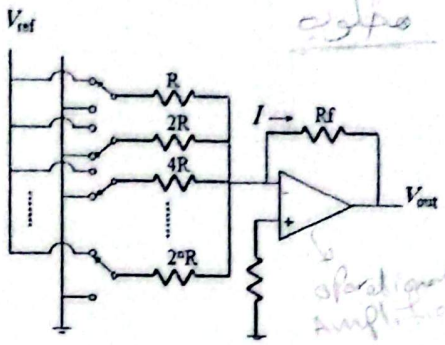
There are two types of DAC and usually used switches, resistors, and op-amps to implement conversion, two types are:

1. Binary Weighted Resistor.
2. R-2R Ladder.

Binary Weighted Resistor:

- Weighted resistors are used to distinguish each bit from the most significant to the least significant.
- Switches are used to switch between Vref and ground (bit high or low).
- Vout= -IRf

- Voltages V1 to Vn are either Vref if corresponding bit is high or ground if corresponding bit is low
- V1 is most significant bit • Vn is least significant bit



$$V_{out} = -IR_f = -R_f\left(\frac{V_1}{R} + \frac{V_2}{2R} + \frac{V_3}{4R} + \cdots \frac{V_n}{2^{n-1}R}\right)$$

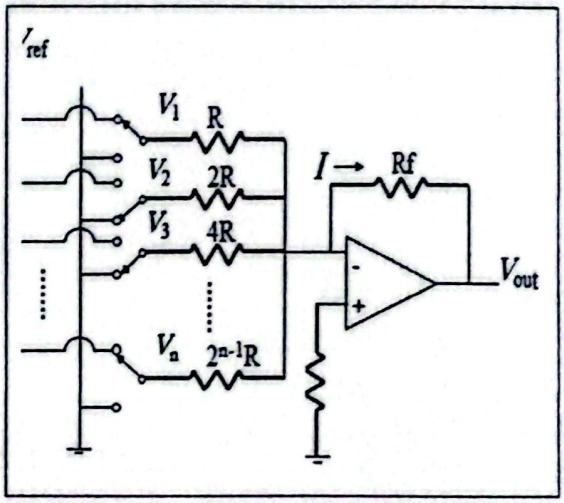
If $R_f = R/2$

$$V_{out} = -IR_f = -\left(\frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{8} + \cdots \frac{V_n}{2^n}\right)$$

For example, a 4-Bit converter yields

$$V_{out} = -V_{ref}\left(b_3\frac{1}{2} + b_2\frac{1}{4} + b_1\frac{1}{8} + b_0\frac{1}{16}\right)$$

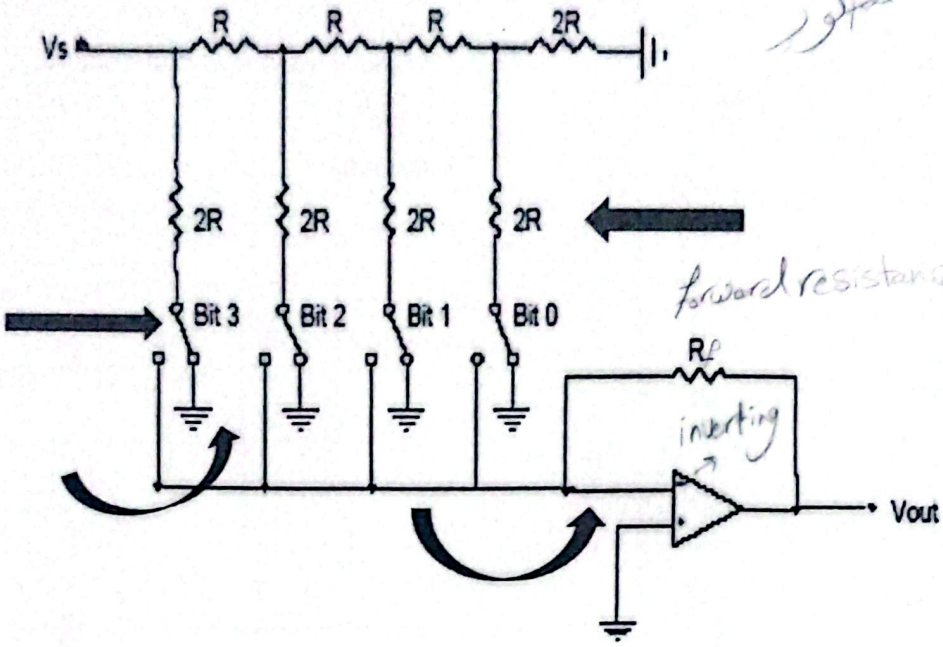
Where b_3 corresponds to Bit-3, b_2 to Bit-2, etc.



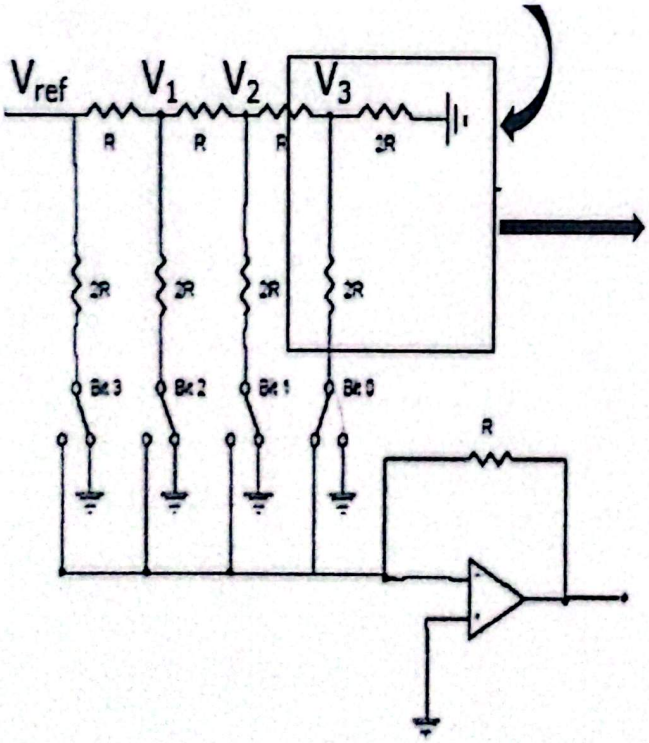
R-2R Ladder:

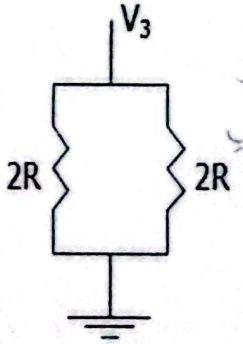
- Each bit corresponds to a switch:
- If the bit is high, the corresponding switch is connected to the inverting input of the op-amp
- If the bit is low, the corresponding switch is connected to ground
- All bits pass through resistance of 2R

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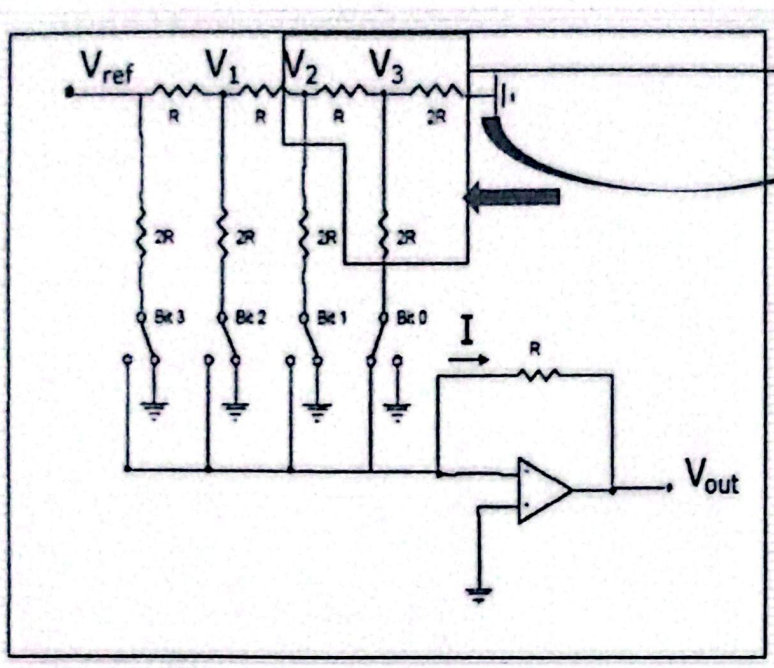
forward resistance

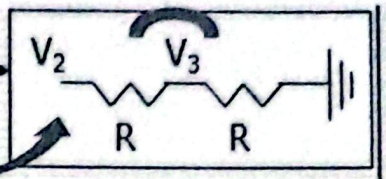




مربوطات على التوازي

$$R_{eq} = \frac{2R * 2R}{2R + 2R} = \frac{4R^2}{4R} = R$$





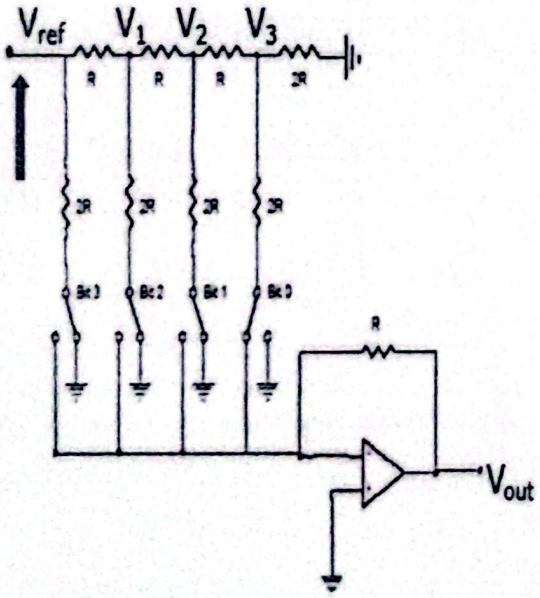
$$V_3 = \left(\frac{R}{R+R}\right)V_2 = \frac{1}{2}V_2$$

Likewise,

$$V_2 = \frac{1}{2}V_1 \longleftarrow$$

$$V_1 = \frac{1}{2}V_{ref} \longleftarrow$$

$$V_{out} = -IR \longleftarrow$$



Home Work

$$V_3 = \frac{1}{8}V_{ref};\ V_2 = \frac{1}{4}V_{ref};\ V_1 = \frac{1}{2}V_{ref}$$

$$V_{out} = -R\left(b_3\frac{V_{ref}}{2R} + b_2\frac{V_{ref}}{4R} + b_1\frac{V_{ref}}{8R} + b_0\frac{V_{ref}}{16R}\right)$$

For a 4-Bit R-2R Ladder

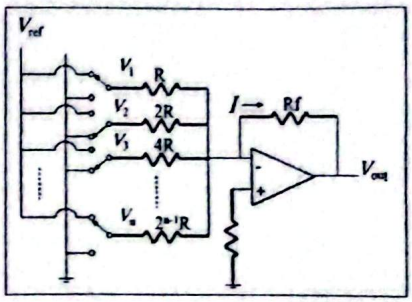
$$V_{out} = -V_{ref}\left(b_3\frac{1}{2} + b_2\frac{1}{4} + b_1\frac{1}{8} + b_0\frac{1}{16}\right)$$

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EX– Derive the equation for Vout for a binary weighted resister with Rf=R/6.

Solution:

$$V_{out} = -IR_f = -R_f\left(\frac{V_1}{R} + \frac{V_2}{2R} + \frac{V_3}{4R} + \cdots \frac{V_n}{2^{n-1}R}\right)$$



if $R_f = R/6$

$$V_{out} = -IR_f = -\frac{R}{6}\left(\frac{V_1}{R} + \frac{V_2}{2R} + \frac{V_3}{4R} + \cdots + \frac{V_n}{2^{n-1}R}\right)$$

$$V_{out} = -\left(\frac{V_1}{6} + \frac{V_2}{12} + \frac{V_3}{24} + \cdots + \frac{V_n}{6 * 2^{n-1}}\right)$$

$V_1 = V_2 = V_3 = V_4 = \ldots = V_n = V_{ref}$

$$V_{out} = -V_{ref}\left(\frac{b_{n-1}}{6} + \frac{b_{n-2}}{12} + \frac{b_{n-3}}{24} + \cdots + \frac{b_{n-n} = b_0}{6 * 2^{n-1}}\right)$$

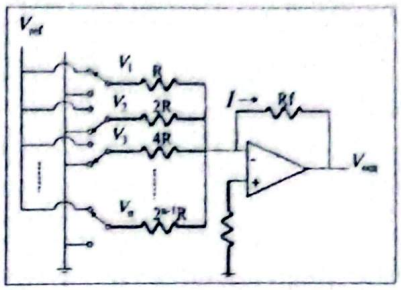
For Example, a 4-bit converter yields:

$$V_{out} = -V_{ref}\left(b_3\frac{1}{6} + b_2\frac{1}{12} + b_1\frac{1}{24} + b_0\frac{1}{48}\right)$$

–EX–Find Vout for a binary weighted resistor D/A with Vref=5 Volts, and codeword = 101111.

Solution:

$$V_{out} = -IR_f = -R_f\left(\frac{V_1}{R} + \frac{V_2}{2R} + \frac{V_3}{4R} + \cdots \frac{V_n}{2^{n-1}R}\right)$$



if $R_f = R/2$

$$V_{out} = -IR_f = -\left(\frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{8} + \cdots \frac{V_n}{2^n}\right)$$

A 6-bit converter yields:

$$V_{out} = -V_{ref}\left(b_5\frac{1}{2} + b_4\frac{1}{4} + b_3\frac{1}{8} + b_2\frac{1}{16} + b_1\frac{1}{32} + b_0\frac{1}{64}\right)$$

$$V_{out} = -5\left(\frac{1}{2} + \frac{0}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64}\right) = -3.671875\ V$$

Questions

1. What is the D/A? why is it needed? Do we always need it?
2. Derive the equation for Vout for a binary weighted resister with Rf=R/6.
3. Derive the equation for Vout for a binary weighted resister with Rf=R/3.
4. Draw the block-diagram of a binary-weighted resistor cct.
5. Find Vout for a binary weighted resistor D/A with Vref=5 Volts, and codeword = 101111.
6. Find Vout for a binary weighted resistor D/A with Vref=3.3 Volts, and codeword = 101101. Find Vout for a binary weighted resistor D/A with Vref=3.3 Volts, and codeword = 101101.
7. Derive the equation of Vout in a R-2R ladder D/A, draw the cct and the minimizations and show the equivalent resistors and voltage markings as well.
8. Find Vout for an R-2R ladder D/A with Vref=3.3 Volts, and codeword = 101101.
9. Find Vout for an R-2R ladder D/A with Vref=3.3 Volts, and codeword = 101101, (assume the switch that is responsible for the LSB is not working and it is stuck in the open state).

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